Yash Goyal

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Work Experience

Argon Robotics June 2026 – Present
Founding Engineering Intern *USA (remote)*

- Owned the **robotics data & observability pipeline**: **concurrent real-time** video feeds, **cloud RTC** to the operator dashboard plus an isolated **out-of-process recorder** with **hardware-accelerated encoding**; RGBD episodes archived to GCS buckets and arm/operator telemetry streamed to **BigQuery**, with **Prometheus + Grafana** dashboards and **25 Slack alert rules**.
- Built **real-time bimanual teleoperation stack** that drives **6-DOF robotic arms** from a Meta Quest, retargeting VR poses through a **differential-IK solver (Pink/Pinocchio)** into a **100 Hz SocketCAN control loop**. Moved video encoding, RGBD recording into **separate OS processes with shared-memory** frame handoff. Hardened the arms for multi-hour sessions with **pose-jump rejection**, dead-man trigger gating, **hold-on-fault fallback** & **sequence-numbered every controller frame**.
- Closed the loop from **teleop data to deployed policy**: **1500+ episodes** cut from **BigQuery telemetry & GCS video**, auto-annotated with a **Gemini VLM** (per-episode phase schema and success/failure labels), exported to **LeRobot datasets on HuggingFace**. **ACT and π0.5 VLA policies** trained and benchmarked across **A100/L4** and deployed on the **PiPER arms**.

Emergent Labs Mar 2026 – May 2026
Forward Deployed Engineering Intern *Bangalore*

- Served as primary technical engineer for **50+ customers** running production apps, diagnosing and resolving Kubernetes (pod crashloops/scheduling), reverse-proxy, and container runtime failures across $\sim$**15 clusters**, cutting average support turnaround $\sim$**45%** and resolving **200+ infra incidents** while keeping high-traffic apps online at **99.9% uptime**.
- Built internal tooling for analytics, data, and ops teams, automating repetitive ETL and reporting workflows with [Airflow/cron jobs], and overhauling observability by instrumenting services with Prometheus metrics, Grafana dashboards, cutting mean-time-to-diagnosis and driving $\sim$**30%** productivity gains across engineering and support.

Successship Technologies Jan 2025 – July 2025
Software Engineering Intern *Bangalore*

- Improved system reliability and cut server costs $\sim$**20%** by triaging critical production issues and optimizing Docker images across **5+ services** through [base-image/multi-stage] build strategies and version upgrades.
- Led end-to-end development of a **distributor management platform** for direct-selling businesses, architecting core inventory (statistical analysis & classification), **ledger/double-entry accounting**, and **payout services** [Spring Boot, Redis, PostgreSQL]; drove the company's **first client signing** and first successful payout through the system.
- Built an AI-powered AP/AR automation system from scratch using [LLM/OCR, LangChain/OpenAI], cutting manual invoice processing by $\sim$**40%**; ran **4+ client demos** that validated usability and shaped early product direction.

Projects

Axon | *NestJS, React, PostgreSQL, Prisma, Redis, Gemini, Docker, GCP*

- Built a **multi-tenant WhatsApp automation SaaS** letting non-technical teams design chatbot conversations on a **drag-and-drop canvas** across **8 node types**, with workspace-scoped **RBAC (4 roles)**, and **immutable published flow versions**
- Engineered a **provider-agnostic conversation engine** serving **Meta Cloud API**, **Twilio**, and a credential-free in-app **sandbox** through one code path, with **HMAC-SHA256/SHA1** webhook signature verification, **Redis-backed redelivery de-duplication**, **24-hour session windows**, and **Gemini fallback** answering off-script queries inside a configured persona.
- Deployed as a **single Docker image** (API + SPA, one port) on **GCP** behind **Caddy**, with **quota metering** before every send, **AES-256-GCM credential encryption**, and **rotating refresh tokens**.

Wanderlust | *AWS EC2, Kubernetes (self-managed), Docker, NodeJS, MongoDB, Redis*

- Deployed a **3-tier cloud-native application** on a **self-managed Kubernetes cluster** provisioned on AWS EC2; orchestrated workloads with Kubernetes Deployments, Services, and ConfigMaps.
- Designed the setup with a **DevOps-first approach**, focusing on infrastructure reliability & container orchestration.

Skills

Programming Languages: Python, Java, JavaScript
Libraries & Frameworks: ReactJS, Node.js, ExpressJS, Spring Boot
Databases: SQL, MongoDB
Cloud & DevOps: CI/CD, Cloudflare, Linux, Nginx, Docker, OTEL, ECR, ECS, Kubernetes
Core CS: Object-Oriented Programming, Data Structures & Algorithms

Education

Rishihood University, Haryana Aug 2023 – Present
Bachelor of Technology (CS & AI) *CGPA: 8.5/10*

Open Source (PR Dashboard)

OWASP Nest

- Enhanced the **frontend/UI** with new features for a better user experience; implemented **Django models** (with test files) for new features to ensure reliability and maintainability; shipped **SlackBot commands** to improve automation and team collaboration; and added **schema validations** while removing dead code for data integrity and a cleaner codebase.

LitmusChaos *(Organization Member)*

- Strengthened **CI quality gates** through **Docker linting** integration, enhanced local setup scripts to **streamline contributor onboarding**, and maintained repository consistency via branding and asset updates.

Palisadoes Foundation *(Organization Member)*

- Stabilized **CI/CD pipelines** with **reusable workflows** and advanced automation, improving build reliability and deployment consistency; led a **large-scale repository migration**; optimized **container images** for faster builds; and implemented production-grade **observability with OpenTelemetry** for improved monitoring and traceability.